

Retention Is Leakage: A Stability–Plasticity–Privacy Trilemma in Federated Continual Learning

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Abstract

- 1 Introduction
- 2 Related Work
- 3 Threat Model and the Artifact Ledger
- 4 Claim C1: Leakage Half-Life Decouples from Accuracy Decay

FIG01 — Retention/leakage decoupling

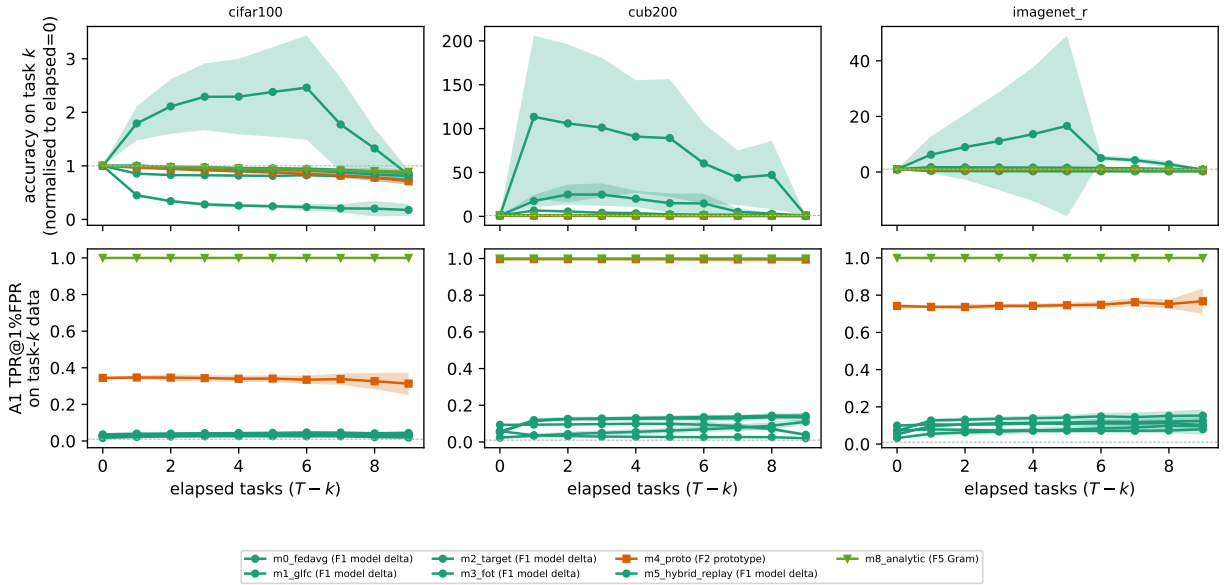


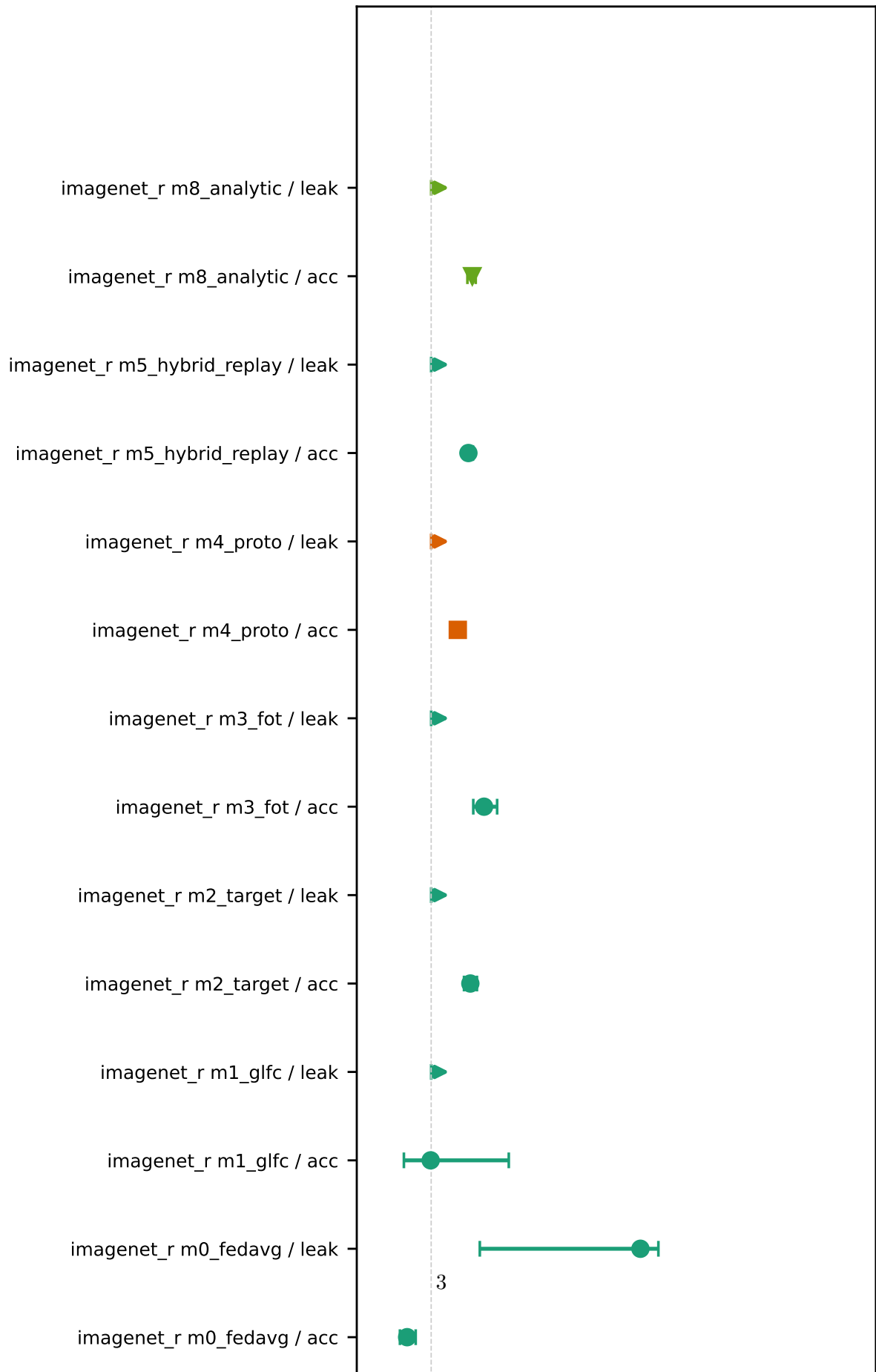
Figure 1: Retention/leakage decoupling, all 7 methods, 3 datasets, 5 seeds.

4.1 Supporting Evidence: Analytic Gram Inversion (H5)

Family	Name	Sufficient-statistic content	A-priori risk	In this zoo
F1	Model deltas	Full gradient signal	High (known); this project's baseline	M0, M1, M2, M3, M
F2	Class prototypes / feature moments	Mean (and covariance) of the client's features for class y	High – a mean over n samples is a linear query; with small n it is nearly a sample	M4 (prototype half)
F3	Prompt pools + keys	Learned input-space perturbations optimized on local data	High (shown centrally exploitable, USENIX'24)	M6 (not built to com
F4	Low-rank adapters	Rank- r update subspace	Medium-High; row space of updates	M4-LoRA (not built
F5	Analytic Gram / autocorrelation	An exact sufficient statistic of the entire local feature matrix (RESEARCH.PLAN.md §3.4)	Critical – see H5's real result: $n=1$ reconstruction is exact	M8
F6	Generative replay / synthetic data	A generative model fitted to client data	Critical (the model *is* the leak)	M2
F7	Class-distribution / count vectors	Exact per-class counts per client per task	Medium alone; high as a join key – deanonymizes and reveals events (A6's H10 test)	M1, M4 (counts)
F8	Exemplar buffers	Literal raw samples	Trivially critical; included as an upper-bound reference point	M1, M5

Table 1: Artifact family taxonomy (F1–F8). See RESEARCH.PLAN.md §2.2 and paper/PAPER.BRIEFING.md §3 for the underlying discussion.

FIG02 — Leakage half-life by family



Dataset	Method	Elapsed	TPR@1%FPR	TPR@0.1%FPR	AUC
cifar100	m0_fedavg	0	0.027 [0.025, 0.030]	0.006 [0.005, 0.008]	0.559 [0.555, 0.563]
cifar100	m0_fedavg	9	0.019 [0.013, 0.024]	0.002 [0.000, 0.005]	0.550 [0.538, 0.562]
cifar100	m1_glf	0	0.016 [0.015, 0.018]	0.002 [0.002, 0.003]	0.521 [0.518, 0.524]
cifar100	m1_glf	9	0.028 [0.015, 0.041]	0.010 [-0.002, 0.022]	0.548 [0.533, 0.563]
cifar100	m2_target	0	0.030 [0.028, 0.032]	0.007 [0.006, 0.008]	0.570 [0.567, 0.573]
cifar100	m2_target	9	0.044 [0.032, 0.056]	0.013 [0.012, 0.014]	0.594 [0.572, 0.615]
cifar100	m3_fot	0	0.036 [0.034, 0.039]	0.009 [0.007, 0.011]	0.589 [0.586, 0.592]
cifar100	m3_fot	9	0.034 [0.026, 0.043]	0.007 [0.003, 0.011]	0.592 [0.576, 0.608]
cifar100	m4_proto	0	0.345 [0.337, 0.352]	0.193 [0.187, 0.200]	0.883 [0.880, 0.886]
cifar100	m4_proto	9	0.314 [0.253, 0.374]	0.166 [0.106, 0.225]	0.869 [0.847, 0.891]
cifar100	m5_hybrid_replay	0	0.028 [0.026, 0.029]	0.006 [0.005, 0.007]	0.566 [0.561, 0.570]
cifar100	m5_hybrid_replay	9	0.036 [0.028, 0.045]	0.015 [0.007, 0.024]	0.566 [0.555, 0.578]
cifar100	m8_analytic	0	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
cifar100	m8_analytic	9	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
cub200	m0_fedavg	0	0.059 [0.056, 0.062]	0.015 [0.014, 0.017]	0.660 [0.656, 0.664]
cub200	m0_fedavg	9	0.020 [0.018, 0.023]	0.003 [0.002, 0.005]	0.602 [0.593, 0.611]
cub200	m1_glf	0	0.024 [0.018, 0.029]	0.002 [0.000, 0.004]	0.561 [0.550, 0.572]
cub200	m1_glf	9	0.110 [0.094, 0.126]	0.043 [0.016, 0.070]	0.681 [0.665, 0.696]
cub200	m2_target	0	0.053 [0.050, 0.056]	0.015 [0.013, 0.017]	0.630 [0.626, 0.634]
cub200	m2_target	9	0.142 [0.127, 0.157]	0.056 [0.040, 0.072]	0.754 [0.743, 0.766]
cub200	m3_fot	0	0.093 [0.090, 0.097]	0.030 [0.027, 0.032]	0.712 [0.708, 0.716]
cub200	m3_fot	9	0.039 [0.032, 0.047]	0.003 [0.002, 0.004]	0.669 [0.657, 0.680]
cub200	m4_proto	0	0.997 [0.995, 0.998]	0.979 [0.970, 0.988]	1.000 [0.999, 1.000]
cub200	m4_proto	9	0.994 [0.986, 1.003]	0.975 [0.946, 1.005]	1.000 [0.999, 1.000]
cub200	m5_hybrid_replay	0	0.055 [0.052, 0.058]	0.015 [0.014, 0.015]	0.648 [0.642, 0.653]
cub200	m5_hybrid_replay	9	0.134 [0.110, 0.158]	0.036 [0.019, 0.053]	0.750 [0.733, 0.767]
cub200	m8_analytic	0	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
cub200	m8_analytic	9	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
imagenet_r	m0_fedavg	0	0.073 [0.067, 0.079]	0.021 [0.017, 0.025]	0.676 [0.670, 0.682]
imagenet_r	m0_fedavg	9	0.080 [0.050, 0.109]	0.012 [-0.008, 0.033]	0.689 [0.667, 0.712]
imagenet_r	m1_glf	0	0.033 [0.029, 0.036]	0.007 [0.005, 0.009]	0.568 [0.559, 0.578]
imagenet_r	m1_glf	9	0.098 [0.071, 0.126]	0.055 [0.033, 0.078]	0.674 [0.656, 0.691]
imagenet_r	m2_target	0	0.052 [0.048, 0.056]	0.013 [0.011, 0.016]	0.631 [0.626, 0.636]
imagenet_r	m2_target	9	0.125 [0.104, 0.146]	0.047 [0.018, 0.077]	0.726 [0.714, 0.738]
imagenet_r	m3_fot	0	0.099 [0.091, 0.108]	0.033 [0.028, 0.038]	0.714 [0.706, 0.721]
imagenet_r	m3_fot	9	0.101 [0.076, 0.126]	0.030 [0.011, 0.048]	0.717 [0.697, 0.738]
imagenet_r	m4_proto	0	0.742 [0.729, 0.754]	0.568 [0.551, 0.585]	0.978 [0.977, 0.980]
imagenet_r	m4_proto	9	0.767 [0.699, 0.836]	0.596 [0.515, 0.677]	0.981 [0.972, 0.990]
imagenet_r	m5_hybrid_replay	0	0.070 [0.065, 0.075]	0.020 [0.017, 0.023]	0.675 [0.671, 0.679]
imagenet_r	m5_hybrid_replay	9	0.153 [0.120, 0.186]	0.089 [0.056, 0.122]	0.737 [0.715, 0.758]
imagenet_r	m8_analytic	0	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
imagenet_r	m8_analytic	9	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]

Table 2: TPR@1%FPR / TPR@0.1%FPR / AUC at elapsed=0 and elapsed=max, per method/dataset.

Method	Family	Quantity	Half-life	CI	R^2	Censored
m0.fedavg	F1	acc	4.68	[3.24, 5.78]	0.771	no
m0.fedavg	F1	leak	17.58	[12.09, 34.64]	0.695	no
m1.glfc	F1	acc	26.20	[15.14, 59.13]	0.043	no
m1.glfc	F1	leak	no decay (> 9)	—	—	yes
m2.target	F1	acc	45.99	[34.55, 67.61]	0.918	no
m2.target	F1	leak	no decay (> 9)	—	—	yes
m3.fot	F1	acc	32.17	[24.41, 41.62]	0.686	no
m3.fot	F1	leak	90.01	[26.15, 614.31]	0.269	no
m4.proto	F2	acc	21.05	[17.78, 25.48]	0.949	no
m4.proto	F2	leak	78.05	[32.81, 791.92]	0.731	no
m5.hybrid_replay	F1	acc	30.52	[22.26, 42.53]	0.955	no
m5.hybrid_replay	F1	leak	no decay (> 9)	—	—	yes
m8.analytic	F5	acc	53.42	[46.55, 59.79]	0.999	no
m8.analytic	F5	leak	no decay (> 9)	—	—	yes
m0.fedavg	F1	acc	5.64	[4.31, 7.72]	0.905	no
m0.fedavg	F1	leak	8.64	[7.71, 9.84]	0.740	no
m1.glfc	F1	acc	5.16	[4.69, 5.83]	0.102	no
m1.glfc	F1	leak	no decay (> 9)	—	—	yes
m2.target	F1	acc	9.31	[6.82, 13.33]	0.015	no
m2.target	F1	leak	no decay (> 9)	—	—	yes
m3.fot	F1	acc	no decay (> 9)	—	—	yes
m3.fot	F1	leak	10.94	[8.85, 14.06]	0.446	no
m4.proto	F2	acc	20.36	[16.42, 27.36]	0.995	no
m4.proto	F2	leak	2737.64	[879.68, 43099.34]	0.827	no
m5.hybrid_replay	F1	acc	5.85	[5.21, 6.96]	0.276	no
m5.hybrid_replay	F1	leak	no decay (> 9)	—	—	yes
m8.analytic	F5	acc	222.54	[176.66, 286.18]	0.998	no
m8.analytic	F5	leak	no decay (> 9)	—	—	yes
m0.fedavg	F1	acc	5.55	[4.80, 6.59]	0.656	no
m0.fedavg	F1	leak	606.48	[23.94, 870.50]	0.007	no
m1.glfc	F1	acc	8.91	[5.20, 42.90]	0.053	no
m1.glfc	F1	leak	no decay (> 9)	—	—	yes
m2.target	F1	acc	19.79	[17.43, 22.60]	0.202	no
m2.target	F1	leak	no decay (> 9)	—	—	yes
m3.fot	F1	acc	26.08	[21.00, 33.93]	0.357	no
m3.fot	F1	leak	no decay (> 9)	—	—	yes
m4.proto	F2	acc	15.38	[13.79, 16.99]	0.992	no
m4.proto	F2	leak	no decay (> 9)	—	—	yes
m5.hybrid_replay	F1	acc	19.04	[17.68, 20.85]	0.485	no
m5.hybrid_replay	F1	leak	no decay (> 9)	—	—	yes
m8.analytic	F5	acc	20.59	[18.66, 22.06]	1.000	no
m8.analytic	F5	leak	no decay (> 9)	—	—	yes

Table 3: Half-life estimates with 95% CI, R^2 , and censoring flag.

FIG13 — Analytic Gram inversion (H5)

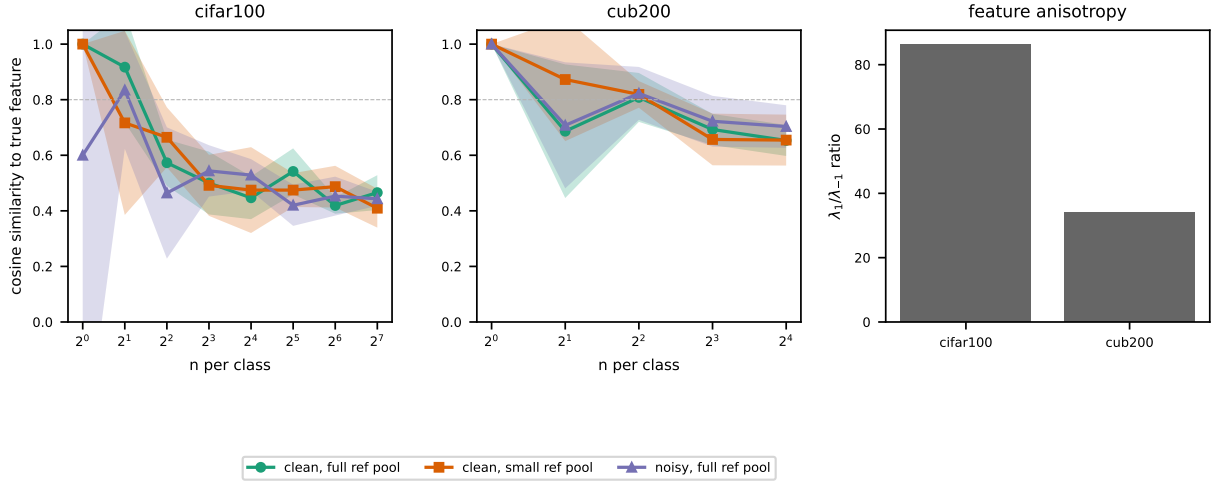


Figure 3: Analytic Gram-inversion cosine similarity vs. per-class n , and feature anisotropy.

4.2 Does Secure Aggregation Help? (H11)

4.3 Is This an Artifact of Synthetic Client Partitions? (FIG18)

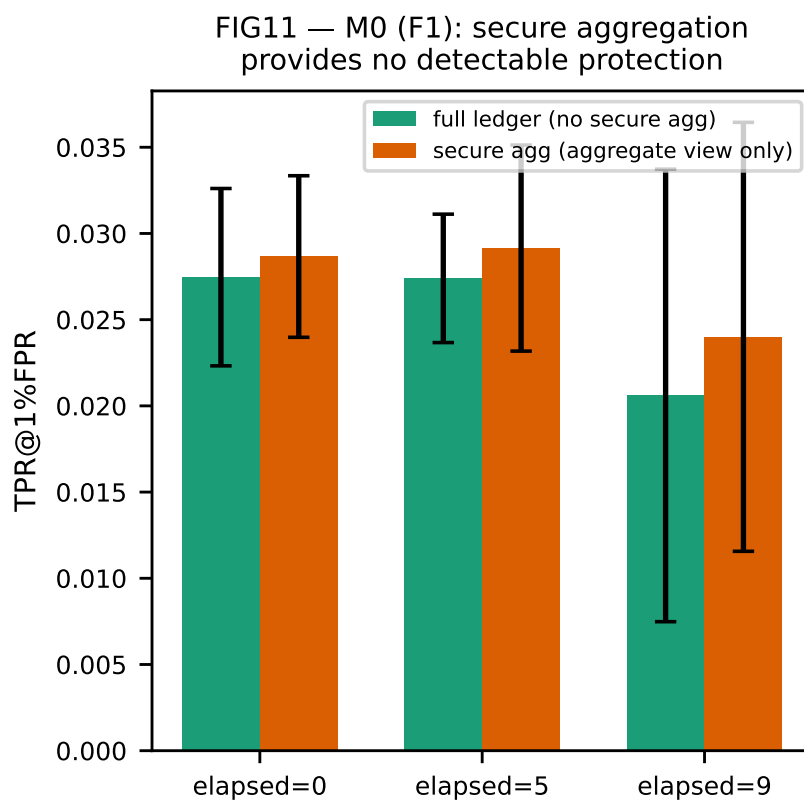
5 Claim C3: Contractive Retention Admits a T -Independent Lifelong ε

6 Claims C2 and C4: Preliminary Evidence and Future Directions

7 Limitations

8 Discussion

References



M4 (F2) and M8 (F5): the attack has no defined score at all under secure aggregation (0/N targets scoreable, every round) -- a structural result, not a bar. See [notes/2026-09-21_secure_agg_a1.md](#).

Figure 4: Secure-aggregation ablation for A1 (M0/F1); M4/M8 (F2/F5) are structurally unattackable (not plotted – no defined score exists to plot).

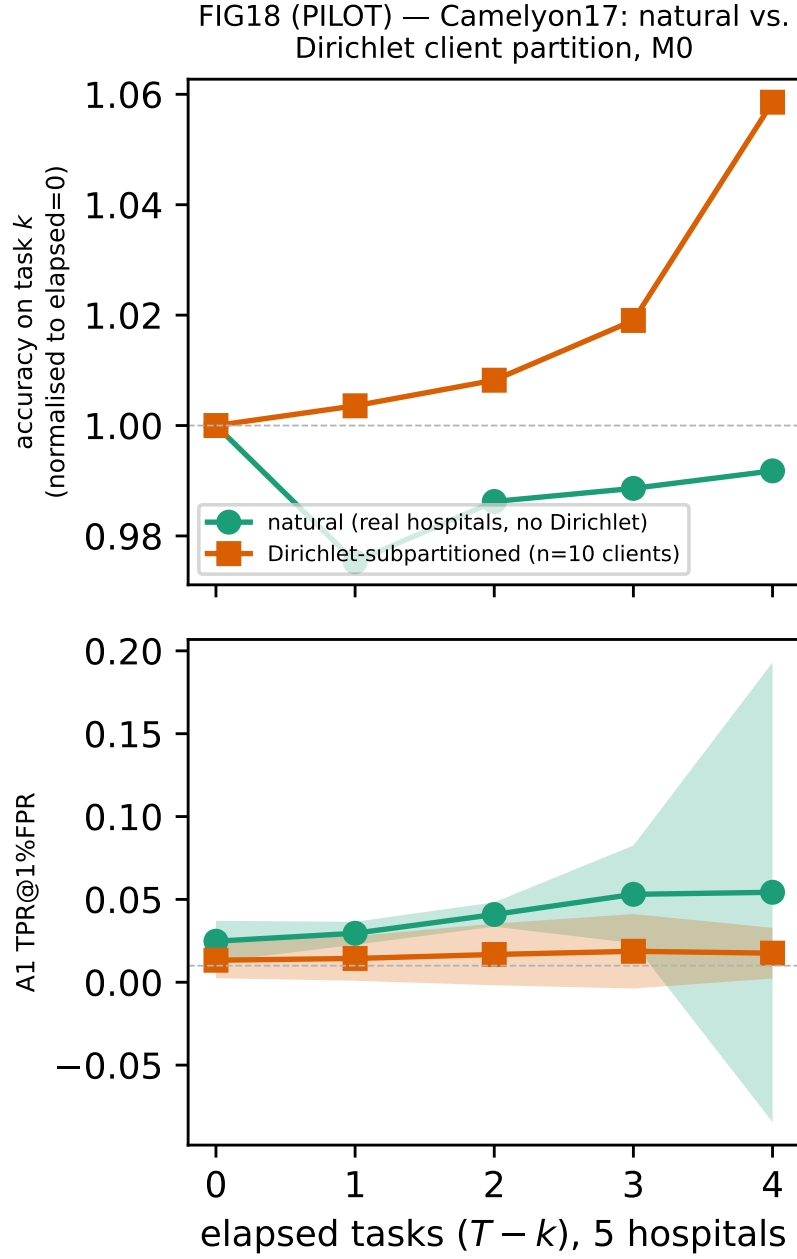


Figure 5: Camelyon17: real hospital partition vs. synthetic Dirichlet partition, M0 (pilot scale).

Unit	Neighbouring relation	Composition regime	Relevance
U1	Example-level (U1): D and D' differ in one example of one client at one task. The weakest unit — what most DP-SGD-in-FL papers silently use.	Sequential composition over all releases that touch the example (no special-casing)	Baseline unit most DP-SGD-in-FL papers use implicitly
U2	Task-level / client-task (U2): D and D' differ in the entirety of one client's data at one task-epoch. Natural for FCL, and the unit under which parallel composition can be legal.	Parallel composition IF the ledger is task-disjoint (flat $\epsilon_{\text{ps}}(T)$); else sequential	The unit under which claim C3 / theorem T1-fwd holds: task-disjoint retention $\leq$ flat $\epsilon_{\text{ps}}(T)$
U3	Client-level, bounded horizon (U3): D and D' differ in all of one client's data across a fixed horizon of T tasks. The standard cross-silo target.	Sequential composition over the fixed horizon T	Standard cross-silo target when a horizon T is fixed in advance
U4	Client-level, unbounded / life-long (U4): D and D' differ in all of one client's data across the unbounded task stream. What the FCL premise implicitly promises; no accountant delivers this non-vacuously for a client who keeps contributing new information forever (H1).	Sequential composition over the *entire* observed stream, unconditionally	The unit under which H1 shows every accountant we tested diverges (theorem T1-conv)
U5	Individual / person-level under renewal (U5): D and D' differ in the data of one person, who may recur across several tasks of one client and possibly across clients, under a renewal (churn) model of the population.	Sequential composition over a person's recurring participation under a renewal model	The unit theorem T2 targets (person-level guarantee under churn), aspirational

Table 4: Units of privacy (U1–U5) and their composition regime.

FIG05 — Lifelong $\varepsilon(T)$ by unit of privacy

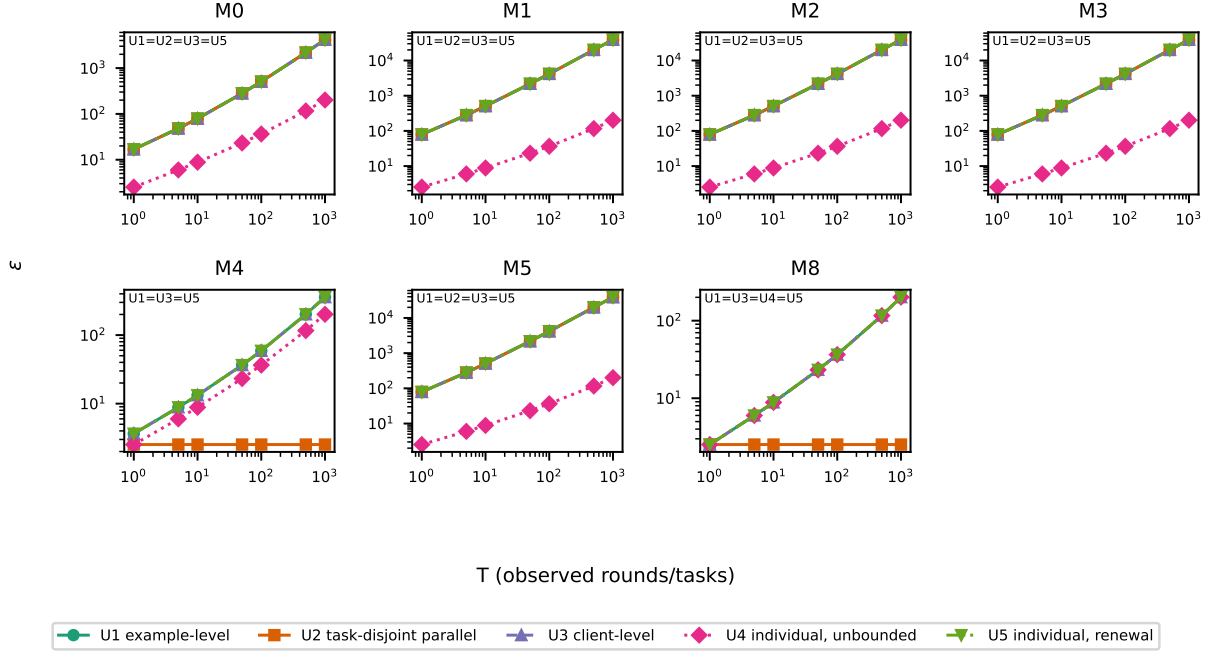


Figure 6: Lifelong $\varepsilon(T)$ by unit of privacy, all 7 methods.

FIG04 (PILOT) — Semantic vs. individual retention

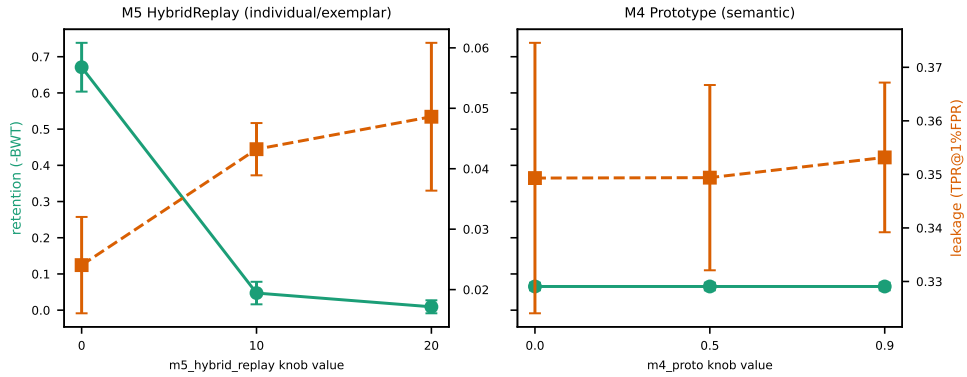


Figure 7: Semantic (M4) vs. individual (M5) retention dose-response pilot.

FIG16 — Log-log ROC, A1 LiRA (elapsed=0, seed=0)

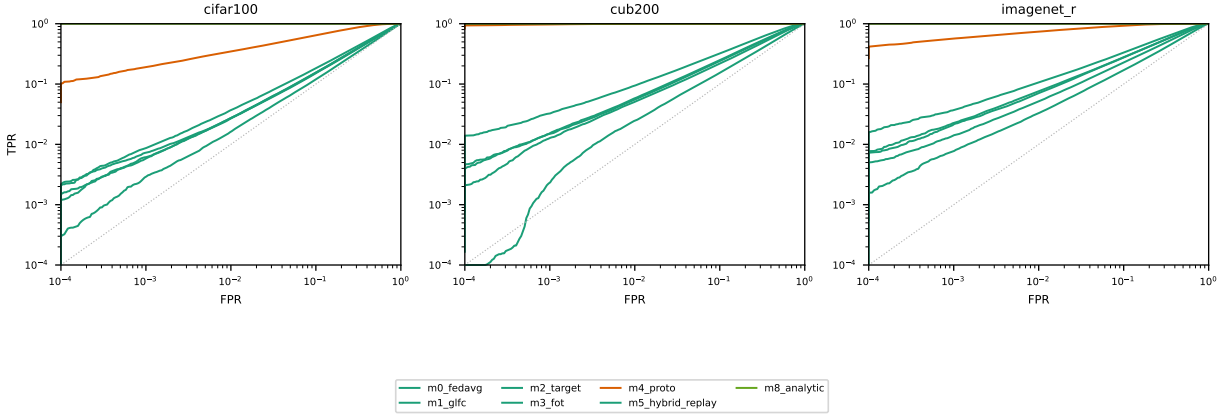


Figure 8: Log-log ROC for A1, all 21 dataset/method combos (appendix candidate).

FIG17 — Per-seed spread of the headline leakage number

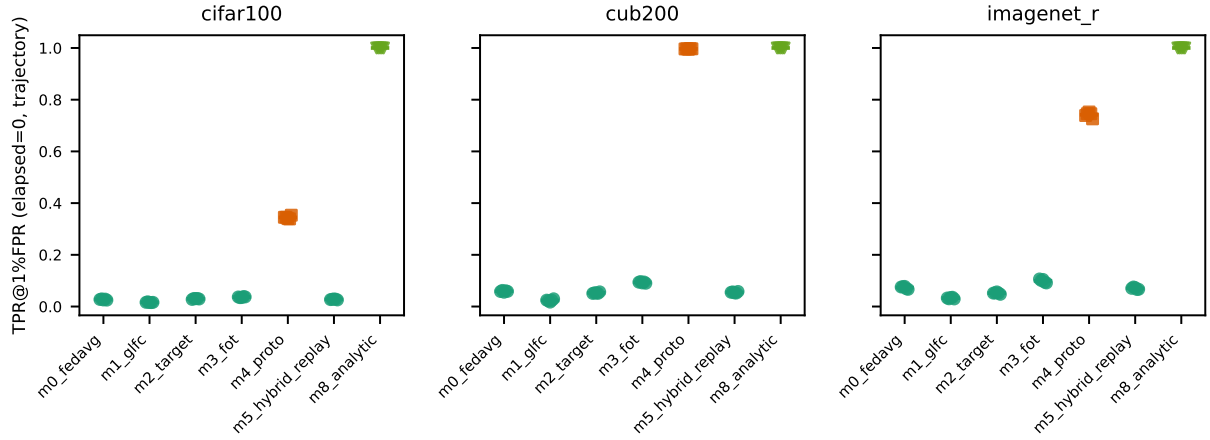


Figure 9: Per-seed spread of the headline leakage number (appendix candidate).